

CHAPTER II.2

Noah Parker

2.1. Let A be a ring, $X = \operatorname{Spec} A$ its spectrum. Fix an element $f \in A$, $D(f) = V(f)^c \subset X$ the associated distinguished open set. Show that $(D(f), \mathcal{O}_X|_{D(f)}) \cong \operatorname{Spec} A_f$ as locally ringed spaces.

Proof. Let $Y = \operatorname{Spec} A_f$, $S = \{f^n\}_{n \geq 1}$, and $\phi : A \rightarrow S^{-1}A = A_f$ be the natural map $a \mapsto a/1$. Note that $f^n \in \mathfrak{p}$ prime for some $n \geq 1$ iff $f \in \mathfrak{p}$ —that is, $S \cap \mathfrak{p} = \emptyset$ iff $\mathfrak{p} \notin D(f)$. Thus, AM 3.11.iv gives us a bijection

$$\begin{aligned} D(f) &\longleftrightarrow Y \\ \mathfrak{p} &\longrightarrow S^{-1}\mathfrak{p}, \\ \phi^{-1}(\mathfrak{q}) &\longleftarrow \mathfrak{q}. \end{aligned}$$

Clearly the map $\mathfrak{q} \mapsto \phi^{-1}(\mathfrak{q})$ is continuous, so it suffices to show that $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is continuous.

Let $V(\mathfrak{b}) \subset A_f$ be closed, and denote $\psi : \mathfrak{p} \mapsto S^{-1}\mathfrak{p}$. AM 3.11.i gives us that $\mathfrak{b} = S^{-1}\mathfrak{a}$ for some ideal $\mathfrak{a} \leq A$. Then $\mathfrak{p} \supset \mathfrak{a}$ implies $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a} = \mathfrak{b}$, so that $V(\mathfrak{a}) \cap D(f) \subset \psi^{-1}(V(\mathfrak{b}))$. Similarly, $S^{-1}\mathfrak{p} \supset S^{-1}\mathfrak{a}$ implies $\phi^{-1}(S^{-1}\mathfrak{p}) \supset \phi^{-1}(S^{-1}\mathfrak{a}) \supset \mathfrak{a}$. We conclude that $\psi^{-1}(V(\mathfrak{b})) = V(\mathfrak{a}) \cap D(f) \subset D(f)$ is closed, so that $\psi = (\psi^*)^{-1}$ is continuous.

By the above, $\phi : A \rightarrow A_f$ induces a map $(g, g^\#) : (\operatorname{Spec} A_f, \mathcal{O}_{\operatorname{Spec} A_f}) \rightarrow (D(f), \mathcal{O}_X|_{D(f)})$ which is a homeomorphism on underlying topological spaces. For each $\mathfrak{q} \not\ni f$,

$$g_{\mathfrak{q}}^\# : \mathcal{O}_{Y, \mathfrak{q}} \cong (A_f)_{\mathfrak{q}} \xrightarrow{\cong} (A_{g(\mathfrak{q})})_f \cong (\mathcal{O}_X|_{D(f)})_{g(\mathfrak{q})}$$

is an isomorphism, so $g^\#$ is an isomorphism of sheaves. Hence, $(g, g^\#)$ is an isomorphism of schemes. $\square$

2.2. Let $(X, \mathcal{O}_X)$ be a scheme, and let $U \subset X$ be any open subset. Show that $(U, \mathcal{O}_X|_U)$ is a scheme. We call this the *induced scheme structure* on the open set U , and we refer to $(U, \mathcal{O}_X|_U)$ as the open subscheme of U .

Proof. Choose a basic cover $\{D(f_i)\}$ of U . Then each

$$(D(f_i), (\mathcal{O}_X|_U)_{D(f_i)}) = (D(f_i), \mathcal{O}_X|_{D(f_i)}) \cong \operatorname{Spec} A_{f_i}$$

by 2.1. We conclude that $\{D(f_i)\}$ is an affine open cover of $(U, \mathcal{O}_X|_U)$, so that the latter locally ringed space is a sheaf. $\square$

2.3. (*) A scheme $(X, \mathcal{O}_X)$ is *reduced* if for every open set $U \subset X$, the ring $\mathcal{O}_X(U)$ has no nilpotent elements.

(a) Show that $(X, \mathcal{O}_X)$ is reduced if and only if for every $x \in X$, the local ring $\mathcal{O}_{X, x}$ has no nilpotent elements.

(b) Let $(X, \mathcal{O}_X)$ be a scheme. Let $(\mathcal{O}_X)_{\operatorname{red}}$ be the sheaf associated to the presheaf $U \mapsto \mathcal{O}_X(U)_{\operatorname{red}}$, where for any ring A , we denote by A_{red} the quotient of A by its nilradical $\mathfrak{R}(A)$. Show that $(X, (\mathcal{O}_X)_{\operatorname{red}})$ is a scheme. We call it the *reduced scheme* associated to X , and denote it by X_{red} . Show that there is a morphism of schemes $X_{\operatorname{red}} \rightarrow X$, which is a homeomorphism on the underlying topological spaces.

- (c) Let $f : X \rightarrow Y$ be a morphism of schemes, and assume that X is reduced. Show that there is a unique morphism $g : X \rightarrow Y_{\text{red}}$ such that f is obtained by composing g with the natural map $Y_{\text{red}} \rightarrow Y$.

Proof. (a) If $(X, \mathcal{O}_X)$ is not reduced, then we can take $U \neq \emptyset$ and $s \in \mathcal{O}_X(U)$ such that $s^n = 0$, whence $s_p^n = 0$ for all $x \in U$. Suppose conversely that there is $x \in X$ such that $\mathcal{O}_{X,x}$ has nilpotent elements. Take such a $(s, U) \in \mathcal{O}_{X,x}$ with $(s, U)^n = 0$. Then there is $V \supset U \supset p$ such that $(s|_V)^n = s^n|_V = 0$.

(b) Consider first the case of $X = \text{Spec } A$ affine. Denote by $(\widetilde{\mathcal{O}_{\text{Spec } A}})_{\text{red}}$ the presheaf $U \mapsto \mathcal{O}_{\text{Spec } A}(U)_{\text{red}}$, so that $(\mathcal{O}_{\text{Spec } A})_{\text{red}} = (((\mathcal{O}_{\text{Spec } A})_{\text{red}})^+)$. By (AM, 3.11), there are isomorphisms $(A_{\text{red}})_f \xrightarrow{\cong} (A_f)_{\text{red}}$ whenever $f \notin \mathfrak{R}(A)$. The same proposition gives us $A_f = 0$ iff $f \in \mathfrak{p}$ for all primes $\mathfrak{p}$, i.e. $f \in \mathfrak{R}(A)$, in which case $(A_{\text{red}})_f = 0$ as well. Then there is an isomorphism of presheaves $\phi : \mathcal{O}_{\text{Spec}(A_{\text{red}})} \rightarrow (\widetilde{\mathcal{O}_{\text{Spec } A}})_{\text{red}}$ given on the distinguished affine opens by $(A_{\text{red}})_f \xrightarrow{\cong} (A_f)_{\text{red}}$. Hence, $(\widetilde{\mathcal{O}_{\text{Spec } A}})_{\text{red}}$ is itself a sheaf, and

$$(\text{Spec } A)_{\text{red}} \cong \text{Spec } A_{\text{red}}$$

is an affine scheme.

(2.3b) gives a natural map of affine schemes

$$(f, f^\#) : \text{Spec } A_{\text{red}} \rightarrow \text{Spec } A$$

induced by the natural projection $\pi : A \twoheadrightarrow A_{\text{red}}$. The map $f : \text{Spec } A_{\text{red}} \rightarrow \text{Spec } A$ of topological spaces is a homeomorphism onto its image $V(\mathfrak{R}(A)) = \text{Spec } A$, as desired.

Now, take X to be an arbitrary scheme, and $\text{Spec } A \cong U \subset X$ an open affine neighborhood. We have

$$\begin{aligned} (U, \mathcal{O}_{X_{\text{red}}}|_U) &\cong (\text{Spec } A, \mathcal{O}_{(\text{Spec } A)_{\text{red}}}) \\ &\cong (\text{Spec}(A_{\text{red}}), \mathcal{O}_{\text{Spec}(A_{\text{red}})}) \end{aligned}$$

is affine, so that X_{red} is a scheme. It suffices to supply a map $X_{\text{red}} \rightarrow X$ which is a homeomorphism on topological spaces.

Take an open cover $\{\text{Spec } A_i \cong U_i \subset X\}_{i \in I}$ of X . The affine case gives us maps $\varphi_i : \text{Spec}(A_i)_{\text{red}} \rightarrow \text{Spec } A_i$. For $i, j \in I$, the affine communication lemma gives a cover of U_{ij} by $D(f_k)$ which are distinguished affine in both A_i, A_j . Since $\mathcal{O}_X(D(f_k)) \cong (A_i)_{f_k} \cong (A_j)_{f_k}$, the restrictions

$$\begin{aligned} \varphi_i|_{D(f_k)} : D(f_k)_{\text{red}} &\rightarrow D(f_k), \\ \varphi_j|_{D(f_k)} : D(f_k)_{\text{red}} &\rightarrow D(f_k) \end{aligned}$$

are induced by the same ring map $\mathcal{O}_X(D(f_k)) \twoheadrightarrow \mathcal{O}_X(D(f_k))_{\text{red}}$, and are hence equal by (2.3). We proceed by making use of the following lemma.

Lemma. *For any schemes $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$, $U \mapsto \text{Hom}_{\mathfrak{S}\text{ch}}((U, \mathcal{O}_X|_U), (Y, \mathcal{O}_Y))$ defines a sheaf of sets.*

Proof. Fix a cover U_i of X , and take maps $(f_i, f_i^\#) : (U_i, \mathcal{O}_X|_{U_i}) \rightarrow (Y, \mathcal{O}_Y)$ such that $(f_i, f_i^\#)|_{U_{ij}} = (f_j, f_j^\#)|_{U_{ij}}$. That is, $f_i|_{U_{ij}} = f_j|_{U_{ij}}$, and the sheaf maps

$$f_i^\#|_{U_{ij}}, f_j^\#|_{U_{ij}} : U_{ij} \rightarrow (f_i|_{U_{ij}})_* \mathcal{O}_Y$$

coincide. It is elementary to show that $U \mapsto \text{Hom}_{\text{Top}}(U, Y)$ is a sheaf of sets, whence the $f_i : U_i \rightarrow Y$ glue to a unique continuous map $f : X \rightarrow Y$. By (Ex. 1.15), $\mathcal{H}om(\mathcal{O}_X, f_*\mathcal{O}_Y)$ is a sheaf, so the $f_i^\# : \mathcal{O}_X|_{U_i} \rightarrow (f_i)_*\mathcal{O}_Y$ glue to a unique sheaf morphism $f^\# : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$. The morphism $f_x^\# : \mathcal{O}_{X,x} \rightarrow (f_*\mathcal{O}_Y)_x$ is determined locally, and is thus a local morphism by locality of the f_i . Hence, our morphisms of locally ringed spaces $(f_i, f_i^\#)$ glue to a unique morphism of locally ringed spaces $(f, f^\#) : (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$, as desired. $\square$

Composing of the inclusion $D(f_k) \hookrightarrow U_{ij}$, uniqueness of gluing implies that $\varphi_i|_{U_{ij}} : (U_{ij})_{\text{red}} \hookrightarrow U_{ij}$ and $\varphi_j|_{U_{ij}} : (U_{ij})_{\text{red}} \hookrightarrow U_{ij}$ coincide. Composing our $\varphi_i : \text{Spec}(A_i)_{\text{red}} \rightarrow \text{Spec } A_i$ with the inclusion $\text{Spec } A_i \hookrightarrow Y$, existence supplies us with our natural sheaf map $\varphi : X_{\text{red}} \rightarrow X$, as desired.

Let $\psi_i := \varphi_i^{-1} : \text{Spec } A_i \rightarrow \text{Spec}(A_i)_{\text{red}}$ be the inverse maps on the underlying topological spaces. Then

$$\psi_i|_{U_{ij}} = (\varphi_i|_{U_{ij}})^{-1} = (\varphi_j|_{U_{ij}})^{-1} = \psi_j|_{U_{ij}}$$

for all $i, j \in I$, so that the ψ_i glue to a continuous map $\psi : \text{Spec } A \rightarrow (\text{Spec } A)_{\text{red}}$. Then

$$(\varphi \circ \psi)|_{U_i} = \psi_i \circ \phi_i = \text{id}_{\text{Spec } A_i}$$

for all $i \in I$ implies that $\varphi \circ \psi = \text{id}_X$, and similarly we conclude $\psi \circ \varphi = \text{id}_{X_{\text{red}}}$. Hence, $\varphi : X_{\text{red}} \rightarrow X$ is a homeomorphism on the underlying topological spaces.

(c) Suppose first that $Y \cong \text{Spec } A$ is affine. Then (Ex. 2.4) gives a bijection between scheme maps $f : X \rightarrow \text{Spec } A$ and ring maps $\varphi := \Gamma(X, f^\#) : A \rightarrow \Gamma(X, \mathcal{O}_X)$. Note that the natural map $\text{Spec } A_{\text{red}} \hookrightarrow \text{Spec } A$ corresponds to the projection $\pi : A \twoheadrightarrow A_{\text{red}}$. X reduced gives us $\varphi(\mathfrak{R}(A)) \subset \mathfrak{R}(\Gamma(X, \mathcal{O}_X)) = 0$, whence the universal property of quotients supplies us with a unique morphism $\psi : A_{\text{red}} \rightarrow \Gamma(X, \mathcal{O}_X)$ such that $\psi \circ \pi = \varphi$. Under the correspondence, this ψ corresponds to a unique map $g : X \rightarrow \text{Spec } A_{\text{red}}$ such that the diagram

$$\begin{array}{ccc} X & \xrightarrow{f} & \text{Spec } A \\ & \searrow g & \uparrow \\ & & \text{Spec } A_{\text{red}} \end{array}$$

commutes, as desired.

Let $f : X \rightarrow Y$, take a cover $\{U_i \cong \text{Spec } A_i\}$ of Y , and let $V_i := f^{-1}(\text{Spec } A_i)$. Then for each $f_i := f|_{V_i} : V_i \rightarrow \text{Spec } A_i$, there is a unique scheme morphism $g_i : V_i \rightarrow \text{Spec}(A_i)_{\text{red}}$ making the following diagram commute.

$$\begin{array}{ccc} V_i & \xrightarrow{f_i} & \text{Spec } A_i \\ & \searrow g_i & \uparrow \\ & & \text{Spec}(A_i)_{\text{red}} \end{array}$$

Cover U_{ij} by distinguished affines $D(a_k)$. The maps $g_i|_{D(f^\#(a_k))}$ and $g_j|_{D(f^\#(a_k))}$ satisfy the given universal property for $f|_{D(f^\#(a_k))} : D(f^\#(a_k)) \rightarrow D(a_k)$, where $D(a_k) \cong \text{Spec}(A_i)_{a_k}$ is affine. Hence, they are equal by the first case.

Composing with our inclusion, we have unique maps $g'_i : V_i \rightarrow \text{Spec}(A_i)_{\text{red}} \hookrightarrow Y_{\text{red}}$ which restrict to the g_i . The proceeding lemma allows us to glue our g'_i to a unique map $g : X \rightarrow Y_{\text{red}}$ such that $g|_{V_i} = g_i : V_i \rightarrow \text{Spec}(A_i)_{\text{red}}$. Let $j_i : \text{Spec}(A_i)_{\text{red}} \hookrightarrow \text{Spec } A_i$ and $j : Y_{\text{red}} \hookrightarrow Y$ be the natural maps. If $g' : X \rightarrow Y_{\text{red}}$ has $f = j \circ g'$, then $f_i = j_i \circ g'|_{V_i}$ implies $g'|_{V_i} = g|_{V_i}$,

whence $g' = g$. We conclude that $g : X \rightarrow Y_{\text{red}}$ is in fact the unique map with this property, as desired. $\square$

2.4. Let A be a ring, and let $(X, \mathcal{O}_X)$ be a scheme. Given a morphism $f : X \rightarrow \text{Spec } A$, we have an associated map $f^\# : \mathcal{O}_{\text{Spec } A} \rightarrow f_* \mathcal{O}_X$ on sheaves. Taking global sections, we obtain a homomorphism $A \rightarrow \Gamma(X, \mathcal{O}_X)$. Thus, there is a natural map

$$\alpha : \text{Hom}_{\mathfrak{S}\text{ch}}(X, \text{Spec } A) \rightarrow \text{Hom}_{\mathfrak{Ring}}(A, \Gamma(X, \mathcal{O}_X)).$$

Show that α is bijective.

Proof. We begin by showing the result for $X = \text{Spec } B$ affine. In this case, α has an inverse map given by

$$\begin{aligned} \beta : \text{Hom}_{\mathfrak{Ring}}(A, B) &\longrightarrow \text{Hom}_{\mathfrak{S}\text{ch}}(\text{Spec } B, \text{Spec } A) \\ \varphi &\longmapsto (f, f^\#), \end{aligned}$$

where

$$\begin{aligned} f(\mathfrak{p}) &= \varphi^{-1}(\mathfrak{p}), \\ f^\#(s)(\mathfrak{p}) &= \varphi_{f(\mathfrak{p})}(s(f(\mathfrak{p}))). \end{aligned}$$

It suffices to show that β and α indeed compose to the identity.

Fix $(f, f^\#) \in \text{Hom}_{\mathfrak{S}\text{ch}}(\text{Spec } B, \text{Spec } A)$, and denote

$$(\beta \circ \alpha)(f, f^\#) = (g, g^\#).$$

Let $\phi_{f(\mathfrak{p})} : A \rightarrow A_{f(\mathfrak{p})}$ and $\phi'_\mathfrak{p} : B \rightarrow B_\mathfrak{p}$ be the natural maps. The commutative square

$$\begin{array}{ccc} A & \xrightarrow{f^\#(\text{Spec } A)} & B \\ \phi_{f(\mathfrak{p})} \downarrow & & \downarrow \phi'_\mathfrak{p} \\ A_{f(\mathfrak{p})} & \xrightarrow{f^\#_{f(\mathfrak{p})}} & B_\mathfrak{p} \end{array}$$

gives us that

$$\begin{aligned} g(\mathfrak{p}) &= (f^\#(\text{Spec } A))^{-1}(\mathfrak{p}) \\ &= (\phi'_\mathfrak{p} \circ f^\#(X))^{-1}(m_\mathfrak{p}) \\ &= (f^\#_{f(\mathfrak{p})} \circ \phi_{f(\mathfrak{p})})^{-1}(m_\mathfrak{p}) \\ &= \phi_{f(\mathfrak{p})}^{-1}(m_{f(\mathfrak{p})}) \\ &= f(\mathfrak{p}), \end{aligned}$$

since $f^\#$ is a local homomorphism. Then

$$\begin{aligned} g^\#(s)(\mathfrak{p}) &= f^\#(X)_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f^\#_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) \\ &= f^\#(s)(\mathfrak{p}). \end{aligned}$$

implies $(g, g^\#) = (f, f^\#)$.

Fix $\varphi : A \rightarrow B$, and denote $\beta(\varphi) = (f, f^\#)$, $(\alpha \circ \beta)(\varphi) = \psi$. Recall that $a \in A$ is identified with $\mathfrak{p} \mapsto a_{\mathfrak{p}} \in A_{\mathfrak{p}}$, and similarly for $b \in B$. Thus, $\psi(a) = \varphi(a)$ iff they're equal as maps $\text{Spec } B \rightarrow \sqcup_{\mathfrak{p} \in \text{Spec } B} B_{\mathfrak{p}}$. The commutative square

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ \downarrow & & \downarrow \\ A_{f(\mathfrak{p})} & \xrightarrow{\varphi_{f(\mathfrak{p})}} & B_{\mathfrak{p}} \end{array}$$

gives us

$$\psi(a)(\mathfrak{p}) = \varphi_{f(\mathfrak{p})}(a(f(\mathfrak{p}))) = \varphi_{f(\mathfrak{p})}(a_{f(\mathfrak{p})}) = \varphi(a)(\mathfrak{p}).$$

Hence, $(\alpha \circ \beta)(\varphi) = \varphi$. We conclude that $\beta = \alpha^{-1}$, so α is a bijection.

Now, we consider the general case of X a scheme. Define a map

$$\begin{aligned} \beta : \text{Hom}_{\mathfrak{H}\text{ing}}(A, \Gamma(X, \mathcal{O}_X)) &\longrightarrow \text{Hom}_{\mathfrak{S}\text{ch}}(X, \text{Spec } A) \\ \varphi &\longmapsto (f, f^\#) \end{aligned}$$

as follows. Fix an affine cover $\{U_i = \text{Spec } B_i\}$ of X , and define $f : X \rightarrow \text{Spec } A$ by $f(x) = \varphi^{-1}(\mathfrak{p}_x)$, where $\mathfrak{p}_x$ is the inverse image in $\Gamma(X, \mathcal{O}_X)$ of the unique maximal ideal $m_x \trianglelefteq (\mathcal{O}_X)_x$. Naturality of the restriction maps gives us that $f|_{U_i} = f_i$, where $(f_i, f_i^\#) = \beta(\rho_{U_i} \circ \varphi)$ is defined in the affine case.

Next, we define $f^\#$ as follows. Fix $V \subset X$ open and $s \in \mathcal{O}_{\text{Spec } A}(V)$. Consider the sections $f_i^\#(V \cap U_i)(s) \in f_*\mathcal{O}_X(V \cap U_i)$, with $f_i^\#$ as above. We have that

$$f_i^\#(V \cap U_i \cap U_j)(s)(\mathfrak{p}) = (\rho_{V \cap U_i \cap U_j} \circ \varphi)_{f(\mathfrak{p})}(s(f(\mathfrak{p}))) = f_j^\#(V \cap U_i \cap U_j)(s)(\mathfrak{p})$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i \cap U_j)$, so the sections agree on their overlap. Then they glue to a unique section $f^\#(V)(s) \in f_*\mathcal{O}_X(V)$.

It suffices to verify that α and β compose to the identity. Fix $(f, f^\#) \in \text{Hom}_{\mathfrak{S}\text{ch}}(X, \text{Spec } A)$, and denote $(\beta \circ \alpha)(f, f^\#) = (g, g^\#)$. Then $g|_{U_i} = g_i$, where

$$(g_i, g_i^\#) = \beta(\rho_{U_i} \circ f^\#(\text{Spec } A)) = \beta(f^\#(U_i)).$$

This gives us $g_i = f|_{U_i}$ by the affine case, so that $g = f$.

Fix $V \subset \text{Spec } A$ open and $s \in \mathcal{O}_{\text{Spec } A}(V)$. Note that the $g^\#(V)(s)$ is the gluing of the sections $g^\#(V \cap U_i)(s)$ defined by

$$\begin{aligned} g^\#(V \cap U_i)(s)(\mathfrak{p}) &= (\rho_{V \cap U_i} \circ f^\#(V))_{g(\mathfrak{p})}(s(g(\mathfrak{p}))) \\ &= f_{f(\mathfrak{p})}^\#(s(f(\mathfrak{p}))) \\ &= f^\#(V \cap U_i)(s)(\mathfrak{p}) \end{aligned}$$

for all $\mathfrak{p} \in f^{-1}(V \cap U_i)$. Thus, $(f, f^\#) = (g, g^\#)$.

Finally, fix $\varphi : A \rightarrow \Gamma(X, \mathcal{O}_X)$ and let $(f, f^\#) = \beta(\varphi)$, $\psi = (\alpha)(f, f^\#)$. Then

$$\rho_{U_i} \circ \psi = \rho_{U_i} \circ (\alpha(f, f^\#)) = f^\#(U_i) = \rho_{U_i} \circ \varphi$$

by the affine case, so that $\psi = \varphi$ by uniqueness of gluing. We conclude that α is a bijection. $\square$

2.5. Describe $\text{Spec } \mathbb{Z}$, and show that it is a final object for the category of schemes, i.e., each scheme X admits a unique morphism to $\text{Spec } \mathbb{Z}$.

Proof. The only primes of $\mathbb{Z}$ are (p) for p prime, and (0) . The former are all maximal, while the latter is a generic point for $\text{Spec } \mathbb{Z}$, since $V(\mathfrak{a}) \ni (0)$ implies $(0) \supset \mathfrak{a}$, i.e. $\mathfrak{a} = (0) \subset \mathfrak{p}$ for all $\mathfrak{p}$ prime.

We have a bijection

$$\text{Hom}_{\mathfrak{S}\text{ch}}(X, \text{Spec } \mathbb{Z}) \rightarrow \text{Hom}_{\mathfrak{R}\text{ing}}(\mathbb{Z}, \Gamma(X, \mathcal{O}_X)),$$

so that the former is singleton for all X by the fact that $\mathbb{Z}$ is initial in $\mathfrak{R}\text{ing}$. $\square$

2.7. Let X be a scheme. For any point $x \in X$, let $\mathcal{O}_x$ be the local ring at x , and m_x its maximal ideal. We define the *residue field* of x on X to be the field $k(x) = \mathcal{O}_x/m_x$. Now, let K be any field. Show that to give a morphism of $\text{Spec } K$ to X , it is equivalent to giving a point $x \in X$ and an inclusion $k(x) \hookrightarrow K$.

Proof. Note that $\text{Spec } K = \{(0)\}$ is singleton, so a morphism $f : \text{Spec } K \rightarrow X$ is a point $x = f((0)) \in X$, as well as the information $f^\#(U) : \mathcal{O}_X(U) \rightarrow K$ for every $x \in U \subset X$ open. There is a single non-empty open set $\{(0)\} = \text{Spec } K \subset \text{Spec } K$, so this gives us a map $f_x^\# : \mathcal{O}_{X,x} \rightarrow K$. Since $f_x^\#$ is a local morphism, we have $f_x^{\#-1}(\mathfrak{m}_K) \subset \mathfrak{m}_{\mathcal{O}_{X,x}}$. Hence, $f_x^\#$ descends to a map $\overline{f_x^\#} : k(x) \rightarrow K$. It is a morphism of fields, and thus an injection.

Conversely, suppose we have an inclusion $k(x) \hookrightarrow K$. Then, for each $x \in U \subset X$, we have a map

$$f^\#(U) : \mathcal{O}(U) \rightarrow \mathcal{O}_{X,x} \twoheadrightarrow k(x) \hookrightarrow K.$$

These maps commute with restriction, thus giving us a morphism $f^\# : \mathcal{O}_X \rightarrow \mathcal{O}_{\text{Spec } K}$ of sheaves. Defining $f : \text{Spec } K \rightarrow X : (0) \mapsto x$ gives us a morphism $(f, f^\#) : \text{Spec } K \rightarrow X$ of ringed spaces. It suffices to show that $f_x^\# : \mathcal{O}_{X,x} \rightarrow K$ is local. We have

$$f_x^{\#-1}(\mathfrak{m}_K) = \ker(\mathcal{O}_{X,x} \twoheadrightarrow k(x)) = \mathfrak{m}_{\mathcal{O}_{X,x}},$$

as desired. $\square$

2.8. (*) Let X be a scheme. For any point $x \in X$, we define the *Zariski tangent space* T_x at $x \in X$ to be the dual of the $k(x)$ -vector space m_x/m_x^2 . Now, assume that X is a scheme over a field k , and let $k[\epsilon]/\epsilon^2$ be the *ring of dual numbers* over k . Show that to give a k -morphism of $\text{Spec } k[\epsilon]/\epsilon^2$ to X is equivalent to giving a point $x \in X$, *rational over* k (i.e. such that $k(x) = k$), and an element of T_x .

2.9. If X is a topological space, and Z an irreducible closed subset of X , a generic point for Z is a point ζ such that $Z = \{\zeta\}^-$. If X is a scheme, show that every (nonempty) irreducible closed subset has a unique generic point.

Proof. We begin by proving the case of $X = \text{Spec } A$ affine. Let $Z = V(\mathfrak{q}) \subset X$ be irreducible and closed. That is, $V(\mathfrak{q}) = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{ab})$ implies $V(\mathfrak{a}) = V(\mathfrak{q})$ or $V(\mathfrak{b}) = V(\mathfrak{q})$. Suppose now that $ab \in \mathfrak{q}$. Then $V((\mathfrak{q}, a)(\mathfrak{q}, b)) = V(\mathfrak{q})$ gives us $(\mathfrak{q}, a) \subset \sqrt{(\mathfrak{q}, a)} = \mathfrak{q}$ or $(\mathfrak{q}, b) \subset \sqrt{(\mathfrak{q}, b)} = \mathfrak{q}$. Hence, either $a \in \mathfrak{q}$ or $b \in \mathfrak{q}$, so that $\mathfrak{q}$ is prime.

Suppose that ζ is a generic point associated to $Z = V(\mathfrak{q})$. Then

$$\{\zeta\}^- = \bigcap_{V(\mathfrak{a}) \ni \zeta} V(\mathfrak{a}) = V(\bigcup_{\mathfrak{a} \subset \zeta} \mathfrak{a}) = V(\zeta)$$

implies that $\zeta = \mathfrak{q}$. This is clearly unique, and we showed that $\mathfrak{q}$ is prime above, so this is always possible.

Let Z be an irreducible closed subset of a scheme X , and let $\{U_i \cong \operatorname{Spec} A_i\}_{i \in I}$ be an affine open cover of X . Replace $\{U_i\}$ by a subcover such that each $U_i \cap Z \neq \emptyset$. We claim that each $U_i \cap Z$ is irreducible in U_i . Take $\emptyset \neq U_i \cap V \subset U_i \cap Z$ nonempty open in U_i . Then $\emptyset \neq U_i \cap V \subset Z$ nonempty open gives us that

$$\operatorname{Cl}_{U_i}(U_i \cap V) = U_i \cap \operatorname{Cl}_X(U_i \cap V) = U_i \cap Z,$$

whence $U_i \cap Z$ is irreducible in U_i .

Take $\zeta_i \in U_i$ to be the unique generic point of $U_i \cap Z$ given by the affine case, so $\{\zeta_i\}^- \supset \operatorname{Cl}_{U_i}(\zeta_i) = Z \cap U_i$. Note that $Z \cap U_i \cap U_j \neq \emptyset$ for all i, j , as otherwise

$$Z = Z \cap (U_i^c \cup U_j^c) = (Z \cap U_i^c) \cup (Z \cap U_j^c)$$

gives us that Z is disjoint from U_i or U_j . Then

$$\{\zeta_i\}^- \cap U_j \supset (Z \cap U_i) \cap U_j \neq \emptyset$$

implies $\zeta_i \in U_j$, as otherwise $U_j^c \cap \{\zeta_i\}^- \supset \{\zeta_i\}^-$. Then $\operatorname{Cl}_{U_j}(\zeta_i) = \{\zeta_i\}^- \cap U_j$ implies

$$\operatorname{Cl}_{U_j}(\zeta_i) \supset (Z \cap U_i \cap U_j)^- \cap U_j = Z \cap U_j.$$

Hence, ζ_i is a generic point of $Z \cap U_j \subset U_j$ for each j . Uniqueness in the affine case gives us $\zeta_i = \zeta_j =: \zeta$ for all i, j , so that

$$\{\zeta\}^- \supset \bigcup_{i \in I} U_i \cap Z = Z.$$

Any generic point ζ' of Z must also be a generic point of each of the $U_i \cap Z$, so uniqueness follows from the affine case. $\square$

2.10. Describe $\operatorname{Spec} \mathbb{R}[x]$. How does its topological space compare to the set $\mathbb{R}$? To $\mathbb{C}$?

Proof. $\square$

2.13. (*) A topological space is *quasi-compact* if every open cover has a finite sub-cover.

- (a) Show that a topological space X is noetherian if and only if every open subset is quasi-compact.
- (b) If A is a noetherian ring, show that $\operatorname{Sp}(\operatorname{Spec} A)$ is a noetherian topological space.
- (c) Give an example to show that $\operatorname{Sp}(\operatorname{Spec} A)$ can be noetherian even when A is not.

Proof. $\square$

- 2.14. (a) Let S be a graded ring. Show that $\operatorname{Proj} S = \emptyset$ if and only if every element of S_+ is nilpotent.
- (b) Let $\varphi : S \rightarrow T$ be a graded homomorphism of graded rings (preserving degrees). Let $U = \{\mathfrak{p} \in \operatorname{Proj} T : \mathfrak{p} \not\supset \varphi(S_+)\}$. Show that U is an open subset of $\operatorname{Proj} T$, and show that φ determines a natural morphism $f : U \rightarrow \operatorname{Proj} S$.
- (c) The morphism f can be an isomorphism even when φ is not. For example, suppose that $\varphi : S_d \rightarrow T_d$ is an isomorphism for all $d \geq d_0$, where d_0 is any integer. Then show that $U = \operatorname{Proj} T$ and the morphism $f : \operatorname{Proj} T \rightarrow \operatorname{Proj} S$ is an isomorphism.

Proof. (a) Denote $\operatorname{nil}' S := \bigcap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over all homogeneous prime ideals of S . It suffices to show that $\operatorname{nil}' S = \operatorname{nil} S$.

For any prime ideal $\mathfrak{p}$, the homogenization $h(\mathfrak{p}) \subset \mathfrak{p}$ generated by its homogeneous elements is clearly a homogeneous prime ideal. Then we have

$$\text{nil}' S \supset \text{nil } S = \bigcap_{\mathfrak{p}} \mathfrak{p} \supset \bigcap_{\mathfrak{p}} h(\mathfrak{p}) = \text{nil}' S,$$

which gives us $\text{nil}' S \subset \text{nil } S$. We conclude that $\text{nil } S = \text{nil}' S$. Now, $\text{Proj } S = \emptyset$ iff $S_+ \subset \text{nil}' S = \text{nil } S$, which is exactly our result.

(b) Let $\mathfrak{p} \in \text{Proj } T$ be a homogeneous prime ideal. Then $\varphi^{-1}(\mathfrak{p})$ is a homogeneous prime ideal, and $\varphi^{-1}(\mathfrak{p}) \supset S_+$ implies

$$\mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(S_+),$$

i.e. $\mathfrak{p} \notin U$. Conversely, $\mathfrak{p} \supset \varphi(S_+)$ implies

$$\varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(S_+)) \supset S_+,$$

so that $\varphi^{-1}(\mathfrak{p}) \notin \text{Proj } S$. We conclude that U is the inverse image of $\text{Proj } S$ under the mapping of homogeneous prime ideals $\mathfrak{p} \mapsto \varphi^{-1}$. Hence, we have a natural map $f : U \rightarrow \text{Proj } S : \mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$.

Fix a homogeneous $g \in T_+$. Then

$$\varphi^{-1}(\mathfrak{p}) \ni g \implies \mathfrak{p} \supset \varphi(\varphi^{-1}(\mathfrak{p})) \ni \varphi(g) \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\varphi(g)) \ni g$$

gives us that

$$f^{-1}(D_+(g)) = D_+(\varphi(g)) \subset U.$$

Hence, $f : U \rightarrow \text{Proj } S$ is continuous. We must now define the map $f^\# : \mathcal{O}_S \rightarrow f_*(\mathcal{O}_T|_U)$ of sheaves.

It suffices to define $f^\#$ on basic open sets. Fix a homogeneous $g \in S_+$, and note that

$$f_*(\mathcal{O}_T|_U)(D_+(g)) = \mathcal{O}_T(f^{-1}(D_+(g)) \cap U) = \mathcal{O}_T(D_+(\varphi(g))) \cong T_{(\varphi(g))}.$$

Recall as that φ is degree preserving, so each of its localizations are as well. In particular, $\varphi_g : S_g \rightarrow T_g$ restricts to a map $\varphi_g : S_{(g)} \rightarrow T_{(\varphi(g))}$. Then the following commutative diagram gives us our sheaf map on $D_+(g)$.

$$\begin{array}{ccc} \mathcal{O}_S(D_+(g)) & \xrightarrow{f^\#(D_+(g))} & f_*(\mathcal{O}_T|_U)(D_+(g)) \\ \downarrow \wr & & \downarrow \wr \\ S_{(g)} & \xrightarrow{\varphi_g} & T_{(\varphi(g))} \end{array}$$

To show that these maps glue together, we note that $D(g) \cap D(h) = D(gh)$, and consider the following commutative diagram.

$$\begin{array}{ccc} S_{(g)} & \xrightarrow{\varphi_g} & S_{(\varphi(g))} \\ & \searrow & \searrow \\ & S_{(gh)} & \xrightarrow{\varphi_{gh}} S_{(\varphi(gh))} \\ & \nearrow & \nearrow \\ S_{(h)} & \xrightarrow{\varphi_h} & S_{(\varphi(h))} \end{array}$$

This defines our sheaf homomorphism on all of $\text{Proj } S$, so that we're done.

(c) Fix d_0 such that $\varphi : S_d \rightarrow T_d$ is an isomorphism for all $d \geq d_0$. We wish to show that $U = \text{Proj } T$, so it suffices to show that $\sqrt{\varphi(S_+)} = T_+$. One inclusion is clear since φ is degree preserving. Surjectivity of $\varphi : S_d \rightarrow T_d$ gives us $\varphi(S_+) \supset \bigoplus_{d \geq d_0} T_d$. Then, for any element $a \in T_+$, a^{d_0} is of degree at least d_0 , so $a^{d_0} \in \varphi(S_+)$ implies $a \in \sqrt{\varphi(S_+)}$. We conclude that $\sqrt{\varphi(S_+)} = T_+$, so $U = \text{Proj } T$.

It suffices to show that $f^\sharp$ is an isomorphism on open sets $D(g)$. Fix homogeneous $g \in S_+$. Since $D(g^n) = D(g)$ for all $n > 0$, we may assume g is of degree $\geq d_0$. We show the map $\varphi_g : S_{(g)} \rightarrow T_{(\varphi(g))}$ is surjective. For any $b/\varphi(g)^n \in (B_A)_{\mathfrak{p}}$, $b\varphi(g) = \varphi(a)$ with $a \in A$ gives us

$$\varphi_g(a/g^{n+1}) = \varphi(ag)/\varphi(g^{n+1}) = b/\varphi(g)^n \in \text{im } \varphi_g.$$

It suffices to check injectivity. If $\varphi(a/g) = 0$, then we can find $n \geq 1$ such that $\varphi(g)^n \varphi(a) = 0$. Then $\varphi(g^n a) = 0$ implies $g^n a = 0$ by injectivity of φ , whence $a/g = 0$ in $S_{(g)}$. The result follows. $\square$

2.16. Let X be a scheme, let $f \in \Gamma(X, \mathcal{O}_X)$, and define

$$X_f := \{x \in X : f_x \notin \mathfrak{m}_x \subset \mathcal{O}_{X,x}\}.$$

- (a) If $U = \text{Spec } B$ is an open affine subscheme of X , and if $\bar{f} \in B = \Gamma(U, \mathcal{O}_X|_U)$ is the restriction of f , show that $U \cap X_f = D(\bar{f})$. Conclude that X_f is an open subset of X .
- (b) Assume that X is quasi-compact. Let $A = \Gamma(X, \mathcal{O}_X)$, and let $a \in A$ be an element whose restriction to X_f is 0. Show that, for some $n > 0$, $f^n a = 0$.
- (c) Now assume that X has a finite cover by open affines U_i such that each intersection $U_i \cap U_j$ is quasi-compact. (This hypothesis is satisfied, for example, if $\text{sp}(X)$ is noetherian.) Let $b \in \Gamma(X_f, \mathcal{O}_{X_f})$. Show that, for some $n > 0$, $f^n b$ is the restriction of an element of A .
- (d) With the hypothesis of (c), conclude that $\Gamma(X_f, \mathcal{O}_{X_f}) \cong A_f$.

Proof. (a) $\mathfrak{p} \in U \cap X_f$ iff $\bar{f}_{\mathfrak{p}} \notin \mathfrak{m}_{\mathfrak{p}} = \mathfrak{p}\mathcal{O}_{U,\mathfrak{p}}$, i.e. $\bar{f} \notin \mathfrak{p}$. It follows that $U \cap X_f = D(\bar{f})$ is open, so that X_f is a union of open sets, hence open itself.

(b) Let $a \in A$ with $a|_{X_f} = 0$. Then, for any open affine U intersection X_f , we define denote the restriction $\bar{a} := a|_U$. Then $\bar{a}|_{D(\bar{f})} = a|_{U \cap X_f} = 0$, whence there exists $n > 0$ such that $\bar{f}^n \bar{a} = \bar{f}^n a = 0$. Quasi-compactness of X gives us a finite affine open cover $\{U_i\}$, so that we can take some fixed $n > 0$ with $(f^n a)|_{U_i} = 0$ for all U_i . The sheaf axiom gives us that in fact $f^n a = 0$, so we are done.

(c) Consider $f_i := f|_{U_i}$ and $b_i := b|_{D(f_i)}$. Then, for each i , (a) gives us some $m > 0$ such that

$$(f_i^m b_i)|_{D(f_i)} = c_i|_{D(f_i)},$$

with $c_i \in \Gamma(U_i, \mathcal{O}_{U_i})$. Denote $U_{ij} = U_i \cap U_j$. $(c_i - c_j)|_{U_{ij} \cap X_f} = 0$, so (c) gives us $l > 0$ such that $f^l(c_i - c_j)|_{U_{ij}} = 0$, i.e. $(f^l c_i)|_{U_{ij}} = (f^l c_j)|_{U_{ij}}$. There are only finitely many U_{ij} , so we can take l such that the above holds for each i, j . Then we have $n = m + l > 0$ such that

$$\begin{aligned} (f_i^n b_i)|_{D(f_i)} &= c_i|_{D(f_i)}, \\ c_i|_{U_{ij}} &= c_j|_{U_{ij}} \end{aligned}$$

for all i, j . Then the c_i glue to an element $c \in A$ such that $c|_{X_f}$ is the gluing of the $f_i^n b_i|_{D(f_i)}$; namely, $f^n b$.

(d) By (c), we have a map $\phi : \Gamma(X_f, \mathcal{O}_{X_f}) \rightarrow A_f : b \mapsto f^n b / f^n$, which is well defined regardless of our choice of $n > 0$.¹ Suppose that $\phi(b) = 0$, so that there exists $m > 0$ with $f^{n+m}b = f^m(f^n b \cdot 1 - f^n \cdot 0) = 0$. Then, with notation as in (c), $f_i^{n+m}b_i = 0$ means $b_i = 0$ for each i , whence $b = 0$, showing injectivity.

Finally, we show surjectivity. Fix $a/f^m \in A_f$. It suffices to find $b \in \Gamma(X_f, \mathcal{O}_{X_f})$ such that $f^{n+m}b - f^n a = 0$ for $f^n b \in A$.² Fortunately, we need only glue together the maps $b_i = a|_{D(f_i)}/f_i^m$ to a map $b \in \Gamma(X_f, \mathcal{O}_{X_f})$, where compatibility is simple. Then $f^m b = a$, so that $\phi(b) = f^m b / f^m = a/f^m$, as desired. We conclude that ϕ is an isomorphism of rings. $\square$

2.17. A Criterion for Affineness

- (a) Let $f : X \rightarrow Y$ be a morphism of schemes, and suppose that Y can be covered by open subsets U_i such that for each i , the induced map $f^{-1}(U_i) \rightarrow U_i$ is an isomorphism. Then f is an isomorphism.
- (b) A scheme X is affine if and only if there is a finite set of elements $f_1, \dots, f_r \in A = \Gamma(X, \mathcal{O}_X)$ such that the open subsets X_{f_i} are affine, and $f_1, \dots, f_r$ generate the unit ideal in A .

Proof. (a) Let

(b) If X is affine, then we can take $(f_1, \dots, f_r) = (1)$ to be a finite set of generators of the unit ideal, so that $X_{f_i} = D(f_i)$ are distinguished affines covering X .

Conversely, take a scheme X with such elements $f_1, \dots, f_r \in A$. Each X_{f_i} is affine and thus quasi-compact, so a finite cover of X by quasi-compact open sets implies X is quasi-compact. Each intersection $X_{f_i} \cap X_{f_j} = D_{X_{f_i}}(f_j) \cong \text{Spec}(A_{f_i})_{f_j} \cong \text{Spec } A_{f_i f_j}$ is affine with and thus quasi-compact, so that 2.16(d) gives us $\Gamma(X_{f_i}, \mathcal{O}_{X_{f_i}}) \cong A_{f_i}$.

Ex. 2.4 supplies us with an isomorphism $\phi_i : X_{f_i} \xrightarrow{\cong} \text{Spec } A_{f_i}$, so that we have a cover of $\text{Spec } A$ by $X_{f_i} \cong \text{Spec } A_{f_i}$ where $\phi_i|_{D_{X_{f_i}}(f_j)} = \phi_j|_{D_{X_{f_i}}(f_j)}$. Then we can glue together the maps ϕ_i to a map $\phi : X \rightarrow \text{Spec } A$, which is an isomorphism by (a). Hence, $X \cong \text{Spec } A$ is affine. $\square$

2.18. In this exercise, we compare some properties of a ring homomorphism to the induced morphism of the spectra of the rings.

- (a) Let A be a ring, $X = \text{Spec } A$, and $f \in A$. Show that f is nilpotent if and only if $D(f)$ is empty.
- (b) Let $\varphi : A \rightarrow B$ be a homomorphism of rings, and let $f : Y = \text{Spec } B \rightarrow X = \text{Spec } A$ be the induced morphism of affine schemes. Show that φ is injective if and only if the map of sheaves $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is injective. Show furthermore in that case f is *dominant*, i.e., $f(Y)$ is dense in X .
- (c) With the same notation, show that if φ is surjective, then f is a homeomorphism of Y onto a closed subset of X , and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective.
- (d) Prove the converse to (c). Namely, if $f : Y \rightarrow X$ is a homeomorphism onto a closed subset, and $f^\# : \mathcal{O}_X \rightarrow f_* \mathcal{O}_Y$ is surjective, then φ is surjective.

Proof. (a) The result is trivial, as $D(f) = \emptyset$ iff $f \in \cap_{\mathfrak{p}} \mathfrak{p} = \text{nil } f$.

¹We are often somewhat sloppy in writing $f^n b$ for the element of A which restricts to it, but we attempt to distinguish when necessary.

²It is in fact equivalent, as if $f^{n+m+l}b - f^{n+l}a = 0$ for $l > 0$, then we can replace $n > 0$ by $n + l > 0$.

(b) In the language of coherent sheaves, we have $\mathcal{O}_X = \widetilde{A}$ and $f_*\mathcal{O}_Y = f_*\widetilde{B} = \widetilde{B_A}$. Then $f^\sharp_{\mathfrak{p}} : \mathcal{O}_{X,\mathfrak{p}} \rightarrow (f_*\mathcal{O}_Y)_{\mathfrak{p}}$ corresponds to the map $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \rightarrow (B_A)_{\mathfrak{p}}$. By CA, we have that φ is injective iff each $\varphi_{\mathfrak{p}}$ is injective, which by the above occurs iff $f^\sharp : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is injective.

It suffices to show that $f(Y)$ is dense in X . Fix $g \in A$. Then

$$\begin{aligned} f(Y) \cap D(g) = \emptyset &\iff f^{-1}(D(g)) = \emptyset \\ &\iff D(\varphi(g)) = \emptyset \\ &\iff \varphi(g) \in \text{nil } B \\ &\iff g \in \text{nil } A \\ &\iff D(g) = \emptyset \end{aligned}$$

by (a) and injectivity of φ respectively. Any nonempty open $U \subset X$ contains a nonempty distinguished open of X , so $f(Y)$ is dense in X .

(c) First, suppose that $\varphi : A \rightarrow B$ is surjective. Localization is exact, so each of the stalks $\varphi_{\mathfrak{p}} : A_{\mathfrak{p}} \rightarrow (B_A)_{\mathfrak{p}}$ is surjective, whence $f^\sharp : \mathcal{O}_X \rightarrow f_*\mathcal{O}_Y$ is surjective. Surjectivity of φ gives us

$$\mathfrak{p} \supset \mathfrak{a} \implies \varphi^{-1}(\mathfrak{p}) \supset \varphi^{-1}(\mathfrak{a}) \implies \mathfrak{p} = \varphi(\varphi^{-1}(\mathfrak{p})) \supset \varphi(\varphi^{-1}(\mathfrak{a})) = \mathfrak{a},$$

so that

$$\begin{aligned} f(V(\mathfrak{a})) &= \{\varphi^{-1}(\mathfrak{p}) \in X : \mathfrak{p} \supset \mathfrak{a}\} \\ &= f(Y) \cap V(\varphi^{-1}(\mathfrak{a})). \end{aligned}$$

In particular,

$$f(Y) = f(Y) \cap V(\ker \varphi) \subset V(\ker \varphi).$$

Take $\mathfrak{q} \in V(\ker \varphi)$, and suppose there exists $a \in \varphi^{-1}(\varphi(\mathfrak{q})) \setminus \mathfrak{q}$. Then there is some $a' \in \mathfrak{q}$ with $\varphi(a) = \varphi(a')$, so that $a - a' \in \ker \varphi \setminus \mathfrak{q} = \emptyset$, a contradiction. Thus, $\varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$.

We now show that $\varphi(\mathfrak{q})$ is prime. Suppose $bb' \in \varphi(\mathfrak{q})$, and take $a, a' \in A$ such that $\varphi(a) = b$ and $\varphi(a') = b'$. Then $aa' \in \varphi^{-1}(\varphi(\mathfrak{q})) = \mathfrak{q}$ implies $a \in \mathfrak{q}$ or $a' \in \mathfrak{q}$. Hence, $b \in \varphi(\mathfrak{q})$ or $b' \in \varphi(\mathfrak{q})$ gives us that $\varphi(\mathfrak{q})$ is prime, so that $g : V(\ker \varphi) \rightarrow Y : \mathfrak{q} \rightarrow \varphi(\mathfrak{q})$ is a set-theoretic inverse to $f : Y \rightarrow V(\ker \varphi) \subset X$. In particular, $f(Y) = V(\ker \varphi)$.

It suffices to show that $g : V(\ker \varphi) \rightarrow Y$ is continuous. We write

$$f(V(\mathfrak{a})) = f(V) \cap V(\varphi^{-1}(\mathfrak{a})) = V(\ker \varphi + \varphi^{-1}(\mathfrak{a})),$$

so that $g^{-1}(V(\mathfrak{a})) = V(\ker \varphi + \varphi^{-1}(\mathfrak{a}))$ is closed in X . We conclude that f is a homeomorphism onto its closed image $f(Y) = V(\ker \varphi)$.

(d) Suppose now that $f : Y \rightarrow X$ is a homeomorphism onto a closed subset, and $f^\sharp : \mathcal{O}_X \rightarrow \mathcal{O}_Y$ is surjective. Let $Z = \text{Spec}(A/\ker \varphi)$, so that the universal property of quotients supplies $\overline{\varphi} : A/\ker \varphi \rightarrow B$ such that $\varphi = \overline{\varphi} \circ \pi$. We denote the associated morphism of sheaves $\overline{f} : Y \rightarrow Z$. Note that $\pi : A \rightarrow A/\ker \varphi$ is surjective, so (c) gives us that the associated map $g : Z \rightarrow V(\ker \varphi) \subset X$ is a homeomorphism.

$\overline{\varphi}$ is injective, so (b) implies $\overline{f}^\sharp : \mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is injective, and $\overline{f}(Y) = f(Y)$ is dense. $f(Y) = f(Y)^-$ is closed, so $\overline{f} : Y \rightarrow Z$ is surjective. $f = g \circ \overline{f}$ with f, g homeomorphisms onto their image implies $\overline{f}$ is a homeomorphism onto its image $\overline{f}(Y) = Z$. Additionally, $f^\sharp = \overline{f}^\sharp g^\sharp$ surjective implies $\overline{f}^\sharp$ is surjective, so that $\overline{f}^\sharp : \mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is an isomorphism of sheaves. Then

$$\varphi = \overline{\varphi} \circ \pi = \overline{f}^\sharp(Z) \circ \pi$$

is the composition of a surjective map with an isomorphism, and is therefore surjective itself. $\square$